National University of Lesotho

Department of Mathematics and Computer Science

CS4433 Software Engineering

Individual Final Project

Moderated Learning Management System

Professional, Legal and Ethical Issues

Student Name: MAKHOTSO RAMOCHELA

Student Number: 202321185

1. Introduction

The Moderated LMS holds the names, enrolments, submissions, marks and feedback of learners, and it decides marks that affect their results. A system like this raises issues beyond the code. This document discusses the professional, legal and ethical issues of the system, says what the design and code do about each one, and says what is still left to the school. Section 2 covers professional issues, Section 3 legal issues, Section 4 ethical issues, Section 5 sums up each issue in one table, and Section 6 lists what I recommend to the school.

2. Professional issues

2.1 Code of conduct

A software engineer has a duty to the public, to the client and to the profession. The ACM/IEEE-CS Software Engineering Code of Ethics (Gotterbarn et al., 1999) says that engineers should act in the public interest, deliver software that meets its requirements and has been tested properly, state limits honestly, and keep the confidentiality of the people whose data they handle. I used these ideas as a checklist for the project:

- **Public interest.** Marks are released only after moderation, so a learner never sees a mark that is still changing. Every mark change by the HoD needs a reason and is written to the audit log.
- **Quality.** The system has 44 unit and integration tests, 103 system tests and 7 acceptance tests. Static analysis, a complexity limit and a code review were used.
- **Confidentiality.** Passwords are stored as Argon2id hashes and the program never shows them. Each role sees only the data it needs.

2.2 Responsibility and competence

The system is small, but a mistake in it can give a learner a wrong mark or show a mark too early. For that reason, the business rules BR-01 to BR-10 are written in one layer and are tested directly, and the user interface cannot skip them (architecture rules AR-1 to AR-10). I also took only the work I could finish and test: six weeks instead of seven, with a short list of Won't-have items (for example, no change of password by the user), which is stated openly in the requirements.

2.3 Maintenance duty

A system stays in use after release, so a professional must make later change safe. This is why every change goes through the change process: a branch, a classification, an impact analysis, new tests and a rerun of the whole suite. CR1 followed this process, and the old tests passed without being changed.

3. Legal issues

3.1 Data protection

The system holds personal data of learners, and many are under 18. In Lesotho the Data Protection Act, 2011 is the law to check (Lesotho, 2011). In general, laws of this kind ask that data be collected for a stated purpose, be limited to what is needed, be kept correct, be kept

safe, not be kept longer than needed, and be open to the person it is about. The table shows how the system compares with these ideas.

Table 3.1: Data protection ideas and the system

Idea	What the system does	What is left to the school
Purpose and limit	A user has only a username, full name, role and password hash. There are no phone numbers, addresses or identity numbers.	State the purpose of the data in a short notice to learners.
Access	The Student sees own work, the Lecturer sees own courses, the HoD and Administrator see only what the role needs. Every service checks the role (AR-4).	Give roles only to the right people and remove them when staff leave.
Accuracy	A mark change keeps the old mark, the new mark, the reason and who made it.	Decide how a learner asks for a correction.
Security	Argon2id hashes, parameterised SQL, lock after five wrong passwords, audit log that the program cannot change.	Protect the computer and the data file (see 3.2).
Keeping data	Users and courses are deactivated and not deleted, so records stay for audit.	Decide how long records are kept and how they are deleted at the end.
Access by the person	Not built. A learner cannot see their own record other than assignments, marks and feedback.	Decide how a request to see the data is answered.

3.2 Security duty

The data file is a single SQLite file and it is not encrypted. A person with access to the computer can copy it or open it with another tool and read or change marks. The program cannot stop this, so the school must limit access to the computer and the file, use disk encryption and keep the backups safe. The backup script copies the file, and the copies need the same protection as the original. A weak or shared password is also a risk, so the school should ask for strong passwords and have at least two Administrators, because a lost password of the only Administrator cannot be recovered.

3.3 Copyright and licences

The system uses four free libraries and tools: SQLite (public domain), libsodium (ISC licence), GoogleTest (BSD 3-clause licence) and CMake (BSD 3-clause licence). These licences allow use in a school system without a fee and ask only that the licence notice is kept. The code written for the project is my own work. Copyright in work produced by a student is also governed by the Copyright and Neighbouring Rights Act of Lesotho and by the rules of the university, and the owner of the code (the student, the university or the school) should be agreed before the system is used outside the course.

3.4 Records and accountability

Marks may be used in appeals and disputes. The audit log records who did each important action and when (for example a mark change, a moderation decision or an extension). This helps the school to answer a complaint with evidence. The log is kept in the same file as the other data, so it has the same weakness as in 3.2, and a copy of the log should be kept in a safe place by the school.

3.5 Contract and warranty

The system was built to the requirements agreed at Gate 1 and tested as described in the reports. It has not been tested by real users or on every computer. A school that uses it should take these limits into account, and the license given by the author should say that the software is provided as it is, without a warranty.

4. Ethical issues

4.1 Privacy of learners

Learners are in a weak position: they must use the system and cannot choose what it records. For this reason, the system collects the smallest amount of data and hides a mark until it has been moderated and released. A learner cannot see another learner's work. An ethical use also means that the school should not use the data for a purpose that learners were not told about, such as ranking people, without a new decision.

4.2 Fairness of the late penalty

The rule comes from the HoD: 10% of the maximum mark for each day late, for up to three days, and zero after that. The system treats everyone in the same way, shows the penalty to the student before they save, and still accepts, marks and gives feedback on late work. Extensions need a reason and record who gave them. The rule is harsh in two places: part of a day counts as a whole day, and the mark falls to zero after 72 hours. These are school policies and I implemented them as given, but I recorded them so that the school can review them. CR1 helps fairness, because a lecturer can give the same extension to a whole class after a shared problem such as a power cut, and does not need to treat students one by one.

One effect of CR1 needs care: a student who enrols after a class extension does not get it, and a student whose work is already marked is skipped. The lecturer is told how many were granted and how many were skipped, so the result is visible, but the lecturer must still handle the skipped cases fairly.

4.3 Power and moderation

The HoD can change a final mark. This power is needed for moderation, but it can be misused. The system limits it: a change needs a reason, the old and new marks are kept, and the change is logged. The Lecturer cannot release marks, because only the HoD releases a moderated mark. These rules support honest decisions, but they do not replace a school policy on who may review the audit log and how often.

4.4 Transparency

A learner should be able to understand why a mark is what it is. The system shows the mark, the penalty and the feedback together, and it shows the original deadline next to an extended one (CR1). The system does not make marking decisions by itself: every mark is entered by a person.

4.5 Accessibility and inclusion

The program uses text only and never shows meaning by color alone, and every function works with the keyboard. The user can change the font and size in the terminal. However, I did not test it with screen readers or with users who have disabilities, and the program is in English

only. Learners with poor sight or who prefer Sesotho may be disadvantaged, so the accessibility is only partly proved.

4.6 Misuse of the system

The system could be misused by a person who has a role: for example, a Lecturer could give extensions to favor a class, or an Administrator could create accounts for people who should not have them. The system cannot know what is right, but it makes such actions visible, because each one is written to the audit log with the name of the person.

5. Summary of the issues

Table 5.1: Issues, how the system deals with them and what is left

Issue	How the system deals with it	Remaining gap
Data protection	Minimum data, role checks, deactivate and not delete	Retention period, notice to learners, access requests
Security	Argon2id, parameterised SQL, lockout, audit log	Unencrypted data file, no password change by the user
Copyright and licences	Free libraries under permissive licences	Agree who owns the code
Professional conduct	Tests, reviews, honest limits, change process	Real-user trials not done
Fairness	Same rule for all, reasons for extensions and mark changes	Review of the part-day and 72-hour rules
Moderation power	Reason, old and new marks and log for every change	School policy for reviewing the log
Accessibility	Text only, keyboard only, no colour-only meaning	No test with screen readers; English only

6. Recommendations to the school

- Write a short privacy notice that tells learners what data is kept, why, and for how long.
- Decide the retention period and how records are deleted at the end.
- Give the data file and the computer to a small group of people, turn on disk encryption and keep backups safe.
- Appoint two Administrators.
- Choose a person to review the audit log each term.
- Review the part-day and 72-hour rules of the late penalty.
- Agree who owns the code and add a license file.
- In a later version: let users change their own password, encrypt the data file, and test with users who have disabilities.

References

Gotterbarn, D., Miller, K. and Rogerson, S. (1999) 'Software Engineering Code of Ethics and Professional Practice (Version 5.2)', ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices.

Lesotho (2011) Data Protection Act, 2011. Maseru: Government of Lesotho.

Lesotho (2011) Copyright and Neighbouring Rights Act, 2011. Maseru: Government of Lesotho.